



Project on

Time-Series Statistical Analysis of Daily Temperature Trends.

Team - StatForge

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1. Introduction

Background & Motivation

- . Temperature and air quality (AQI) are key indicators of environmental conditions.
- . Rapid urbanization has led to increasing pollution levels in cities like Delhi.
- . Understanding patterns in temperature and AQI helps in environmental monitoring.
- . Time-series analysis enables identification of trends, variations, and anomalies.
- . Statistical techniques help convert raw data into meaningful insights

Problem Context

- . Environmental data shows continuous fluctuations over time.
- . Manual analysis of such data is difficult and inefficient.
- . Lack of clear insights into relationship between temperature and AQI
- . Need for statistical methods to identify trends, variability, and correlations
- . Requirement of predictive analysis for better environmental understanding

2. Problem Statement

- . Environmental parameters like temperature and AQI vary continuously over time
- . Raw data does not directly provide meaningful insights
- . Lack of clear understanding of relationship between temperature and air quality

Core Gap

- . Absence of structured time-series analysis on environmental data
- . Limited use of statistical techniques to analyze variability and trends
- . No clear visualization of relationships between temperature and AQI
- . Need for a systematic approach to extract actionable insights
- . Inability to detect anomalies and fluctuations effectively

3. Objectives

- . To analyze temperature and AQI data using statistical techniques
- . To identify trends and patterns through time-series analysis
- . To apply moving averages for smoothing and trend detection
- . To examine the relationship between temperature and AQI using correlation analysis
- . To detect anomalies and variations in environmental data
- - To visualize data using graphs for better interpretation and insights

4. Literature Review

Author / Study	Method Used	Key Findings
Study 1	Time-Series Analysis	Identified temperature trends over time
Study 2	Statistical Analysis	Used mean and standard deviation to analyze variability
Study 3	Correlation Analysis	Found relationship between temperature and air quality
Study 4	Moving Average	Smoothed fluctuations to identify patterns
Study 5	Data Visualization	Graphical methods improved interpretation of data

5. Dataset Description

- The dataset consists of daily environmental data collected for Delhi over a period of 90 days
- It includes key variables such as Temperature, Air Quality Index (AQI), and Humidity
- Data is structured in time-series format with Date as the primary index
- Temperature is measured in degrees Celsius ($^{\circ}\text{C}$), AQI represents air quality levels, and Humidity is given in percentage (%)
- The dataset is clean, continuous, and suitable for statistical and time-series analysis
- It enables analysis of trends, patterns, correlations, and anomalies in environmental conditions

6. Methodology

- Data Collection: Environmental data (Temperature, AQI, Humidity) collected for a continuous time period
- Data Preprocessing: Cleaning and formatting of data, handling missing values, and converting date into proper time-series format
- Descriptive Statistics: Calculation of mean, median, standard deviation, and variance to understand data distribution
- Time-Series Analysis: Visualization of temperature and AQI trends over time using line graphs
- Moving Average: Applied to smooth fluctuations and identify underlying trends
- Correlation Analysis: Examined relationship between temperature and AQI
- Outlier Detection: Used statistical methods (Z-score) to identify anomalies
- Visualization: Graphical representation using plots such as line charts, histograms, heatmaps, and scatter plots
- Trend Analysis: Applied regression techniques to identify overall trend patterns

7. SWOT Analysis

Strengths

- . Uses real-world environmental dataset.
- . Combines statistical and time-series analysis techniques

Weaknesses

- . Limited dataset size (short time period)
- - Does not include external influencing factors (traffic, industry, etc.)

Opportunities

- . Can include more variables like wind speed and rainfall
- - Useful for environmental monitoring and decision-making

Threats

- . Incomplete or inaccurate data may affect results
- - Sudden environmental changes may impact analysis

Expected Outcomes

- . Identification of trends and patterns in temperature and AQI
- - Detection of anomalies and fluctuations

8. Results and discussion

◆ Key Results

Temperature shows a gradual increasing trend over the observed period
AQI shows a decreasing pattern as temperature increases
Moving average smooths short-term fluctuations and highlights underlying trends
Correlation analysis indicates a negative relationship between temperature and AQI
Rolling statistics reveal variability in temperature over time
Outlier detection identifies a few abnormal temperature values

◆ Discussion

Higher temperatures are associated with improved air quality (lower AQI values)
Time-series analysis effectively captures trends and seasonal variations
Visualization techniques help in understanding complex data patterns
Statistical measures provide insights into variability and consistency

◆ Evaluation Criteria (KPIs)

Accuracy of trend detection using time-series analysis
Strength of correlation between temperature and AQI
Effectiveness of moving average in smoothing data
Ability to identify anomalies using statistical methods
Clarity and interpretability of visualizations